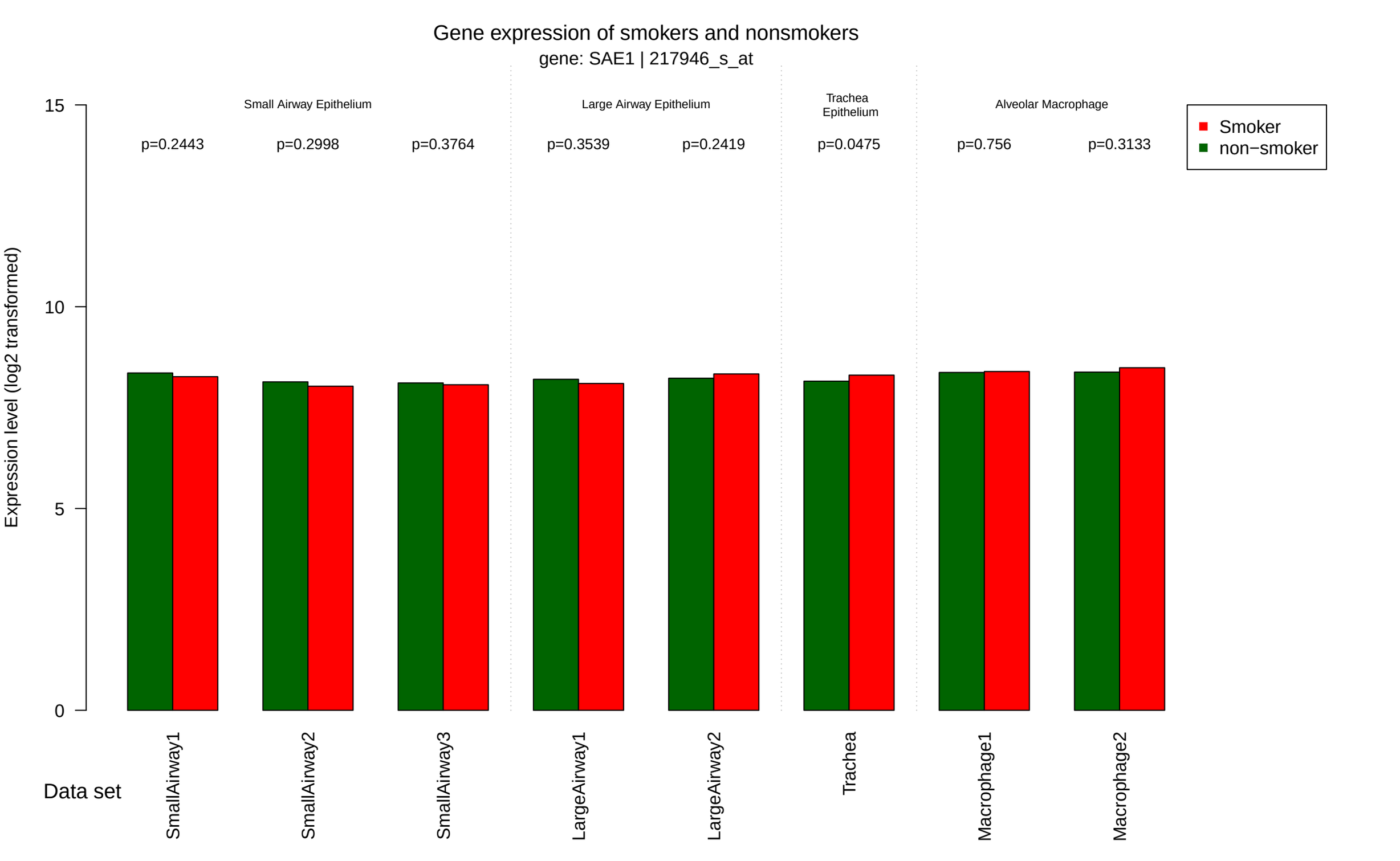
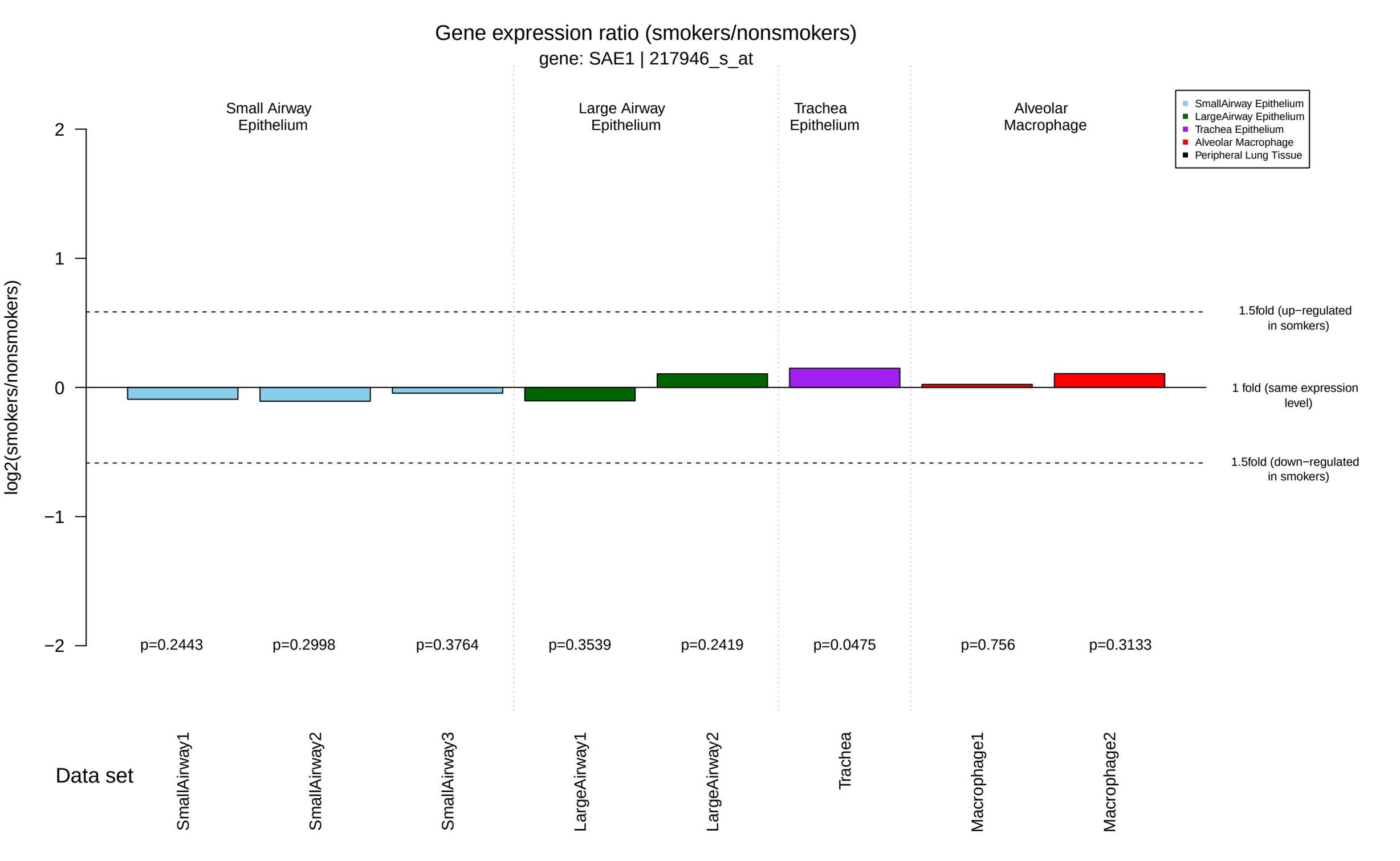
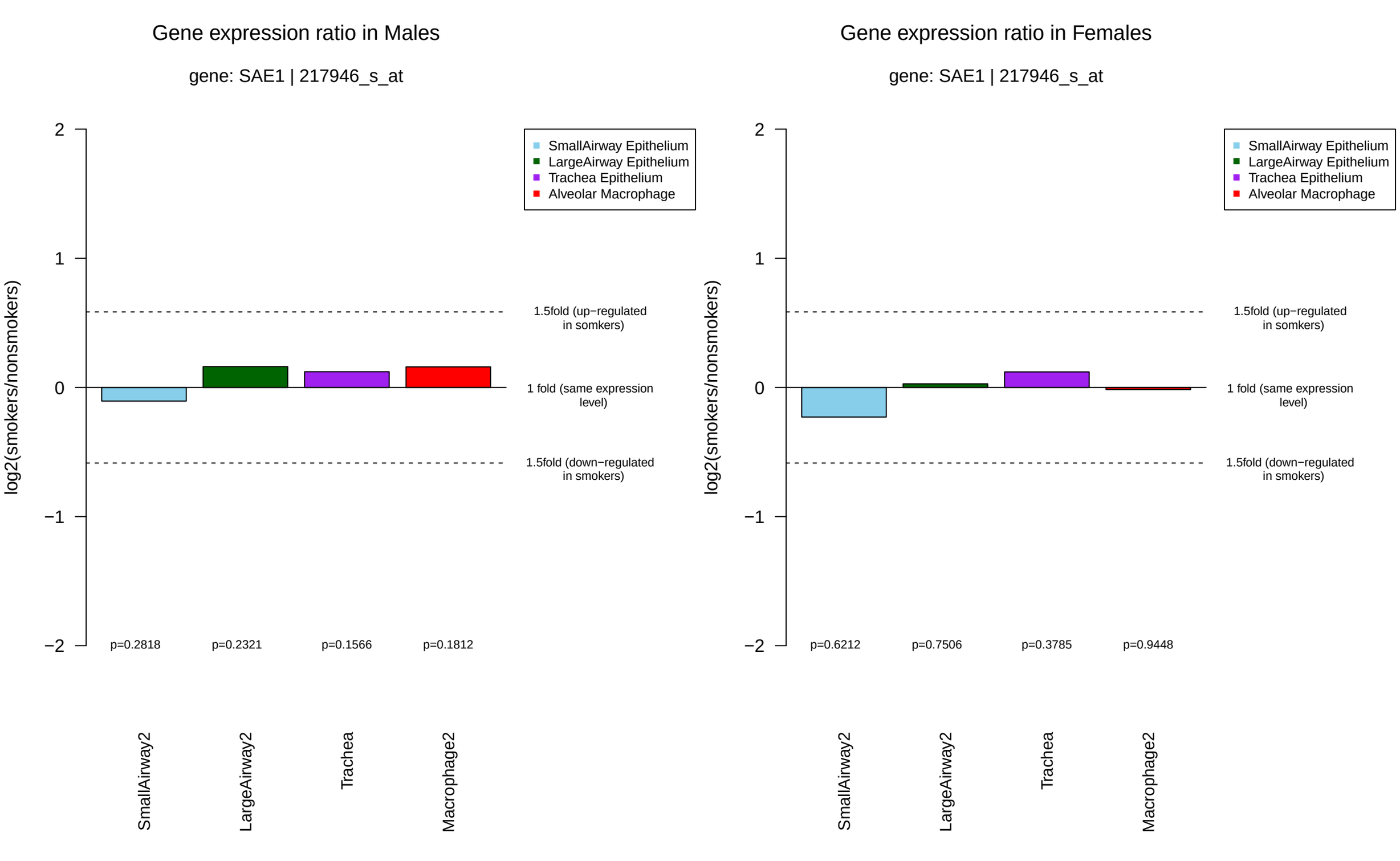
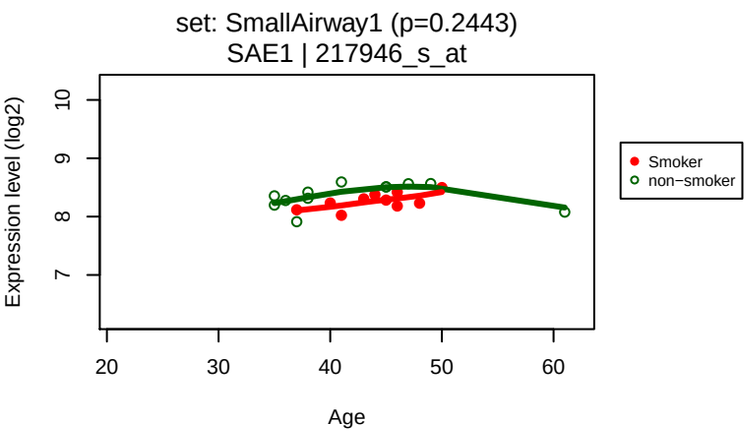
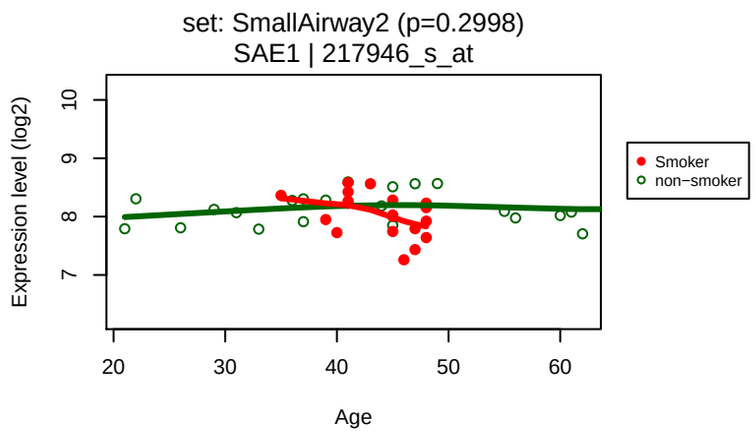
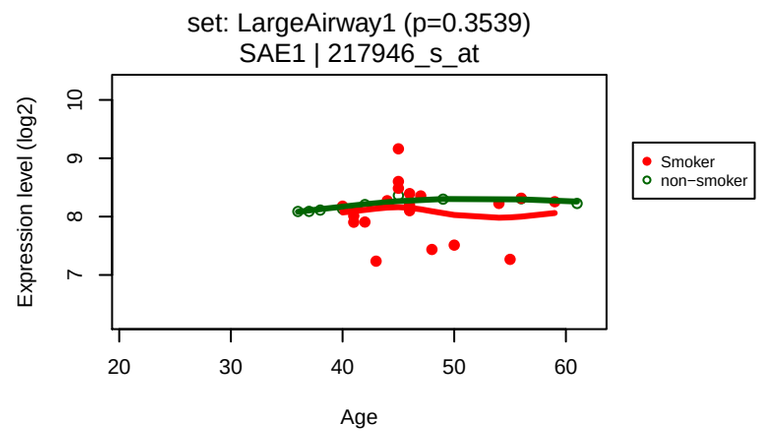

Gene expression of smokers and nonsmokers
gene: SAE1 | 217946_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2443
p=0.2998
p=0.3764
p=0.3539
p=0.2419
p=0.0475
p=0.756
p=0.3133
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

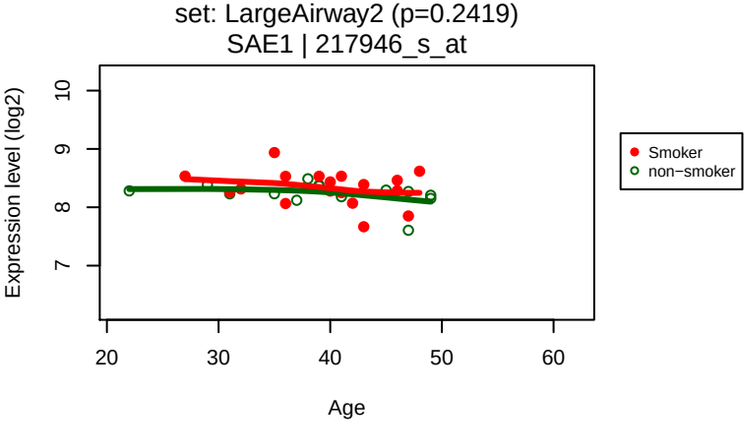

Gene expression ratio (smokers/nonsmokers)
gene: SAE1 | 217946_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2443
p=0.2998
p=0.3764
p=0.3539
p=0.2419
p=0.0475
p=0.756
p=0.3133
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

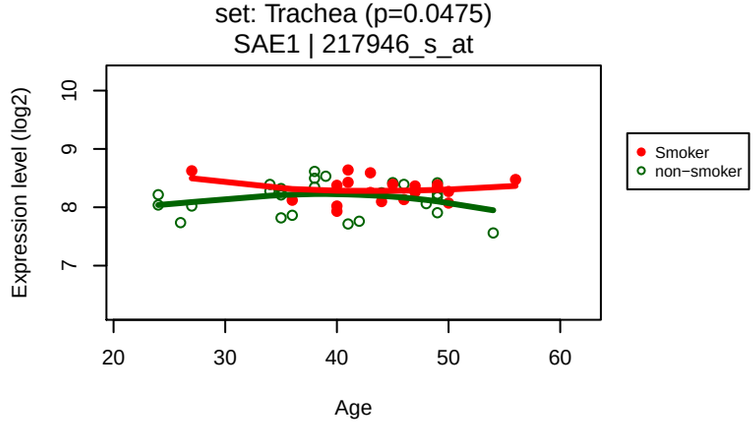

Gene expression ratio in Males
gene: SAE1 | 217946_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2818
p=0.2321
p=0.1566
p=0.1812
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SAE1 | 217946_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6212
p=0.7506
p=0.3785
p=0.9448
SmallAirway2
LargeAirway2
Trachea
Macrophage2

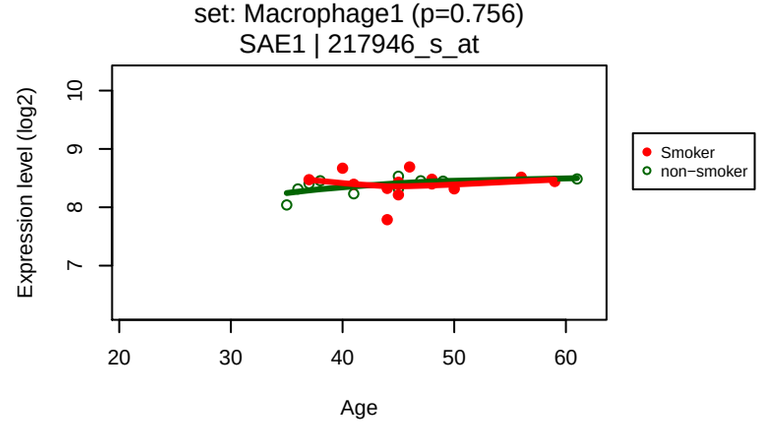

set: SmallAirway1 (p=0.2443)
SAE1 | 217946_s_at
Expression level (log2)
Age
Smoker
non−smoker

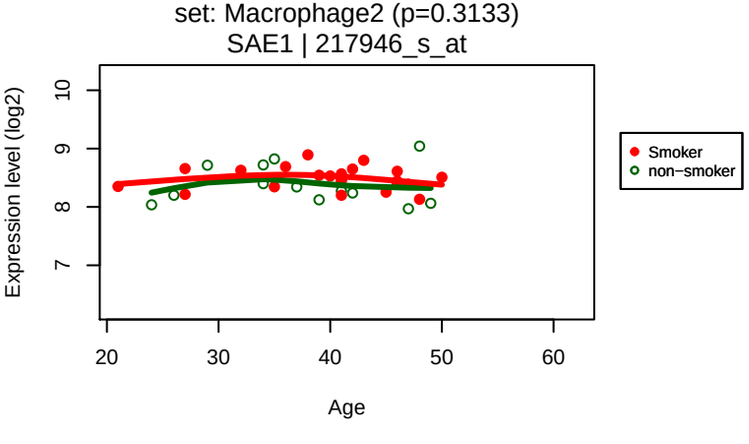

set: SmallAirway2 (p=0.2998)
SAE1 | 217946_s_at
Expression level (log2)
Age
Smoker
non−smoker

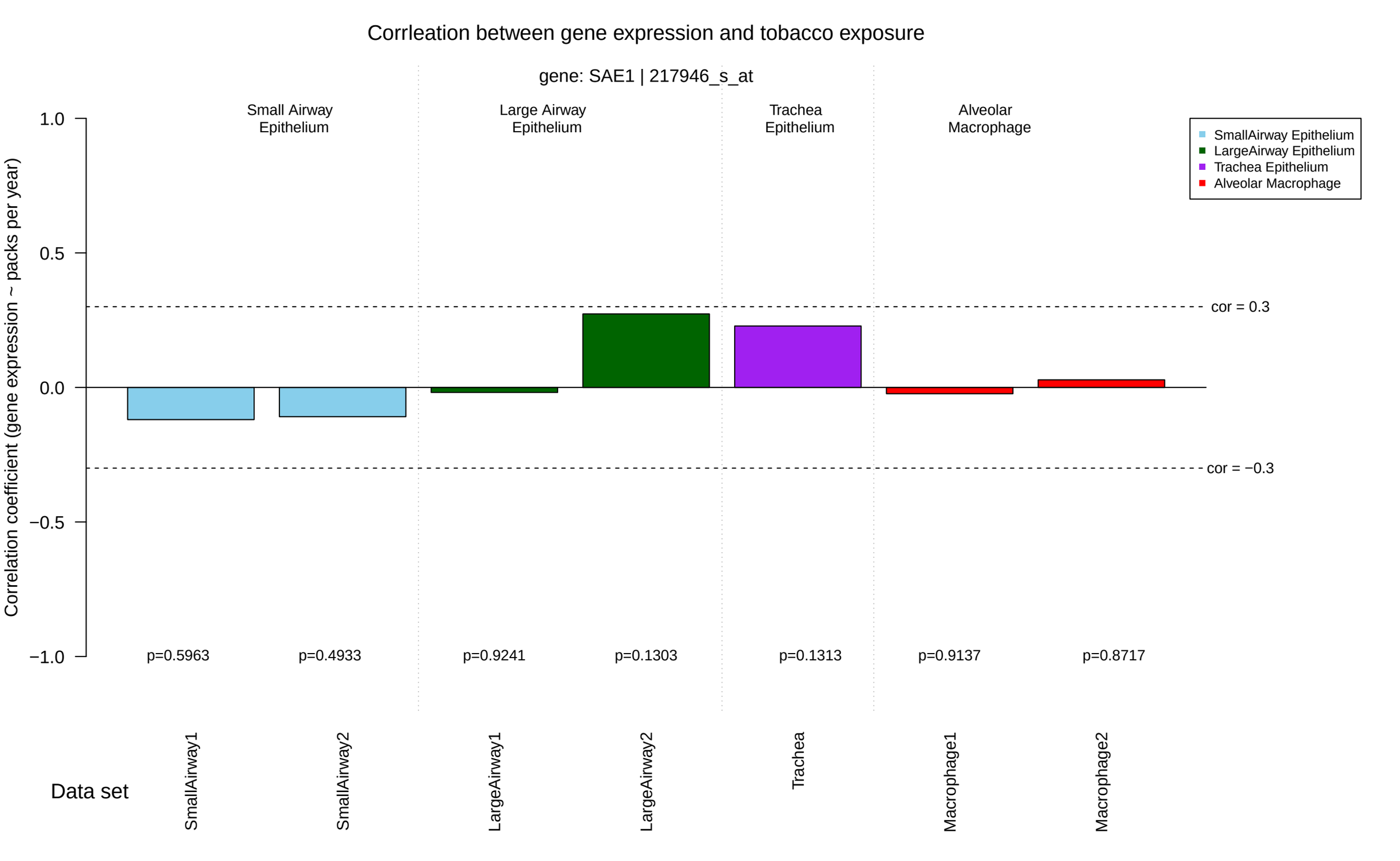

set: LargeAirway1 (p=0.3539)
SAE1 | 217946_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2419)
SAE1 | 217946_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0475)
SAE1 | 217946_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.756)
SAE1 | 217946_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3133)
SAE1 | 217946_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SAE1 | 217946_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5963
p=0.4933
p=0.9241
p=0.1303
p=0.1313
p=0.9137
p=0.8717
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2